MiniProject1

Ro Neel

The Questions

- How do word frequencies between two works of a single author compare?
- How do word frequencies between works of two authors compare?

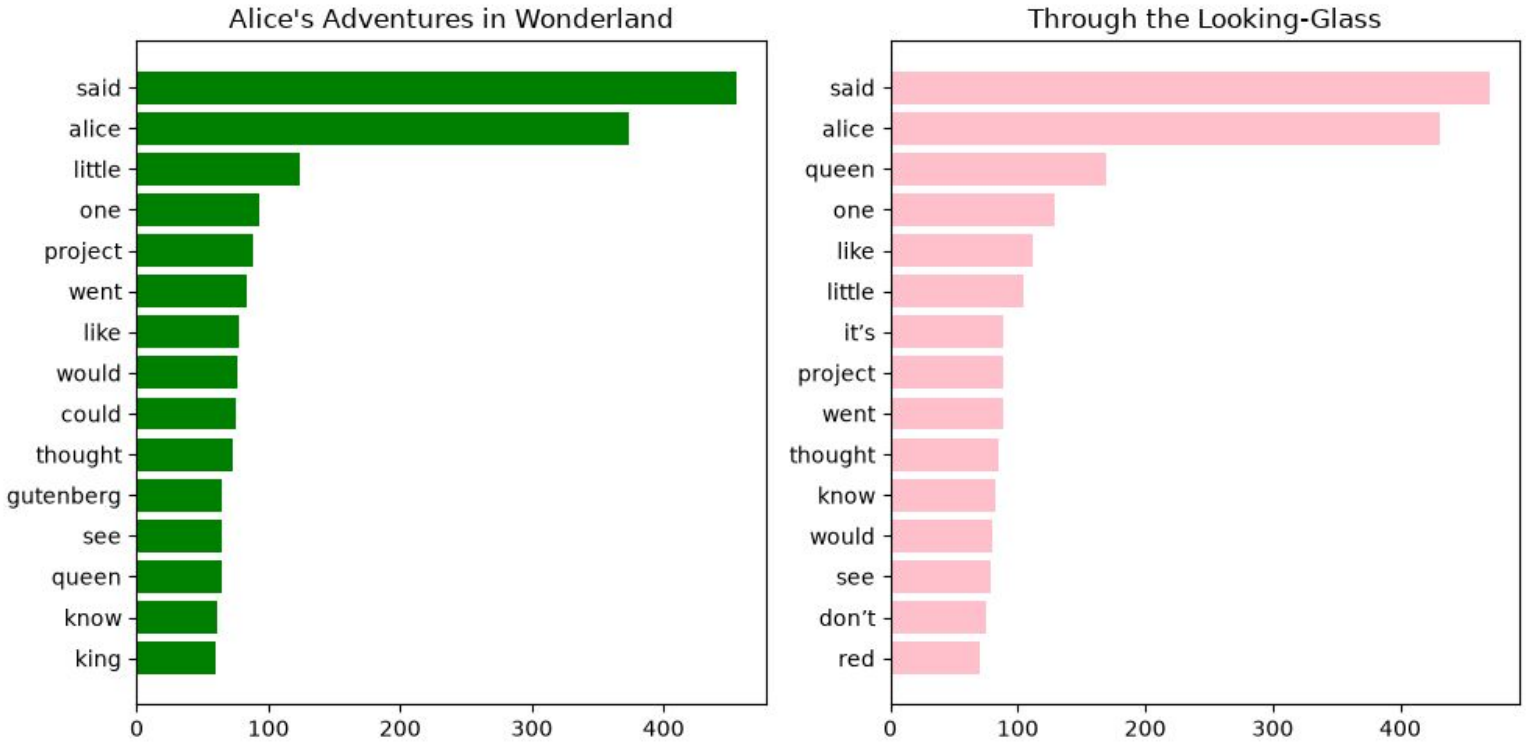
The Approach

- Using Python, pulled two works by Lewis Carroll and one by Bram Stoker from Project Gutenberg
- Cleaned HTML, split words by white space, and cleaned each word of trailing characters
- Counted each occurrence of the words, stored in a tuple list and removed stop words
- Sorted by highest count and compared

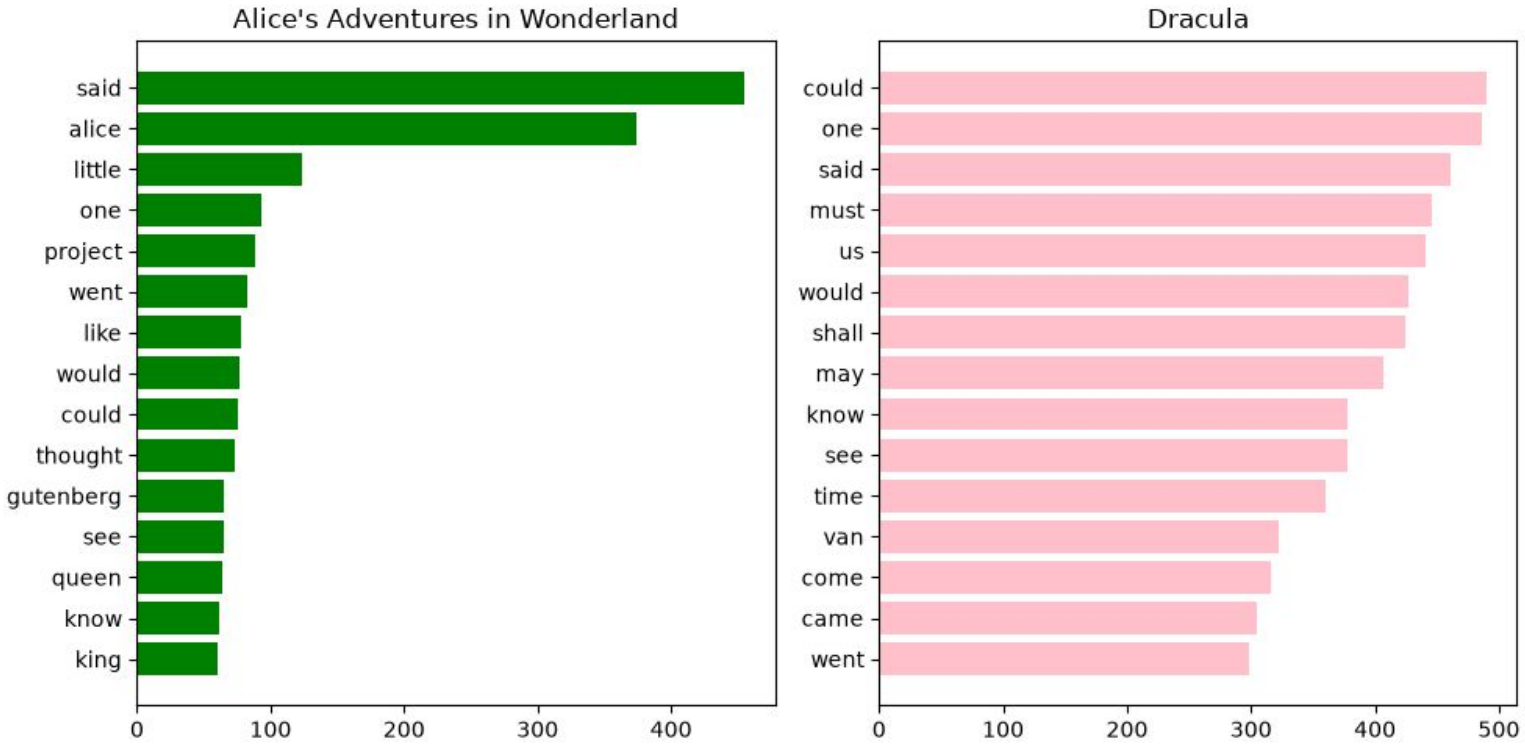
Problems Encountered

- Some “words” were empty and incorrect filtering needed to be fixed
- Figuring out bar graphs with matplotlib was difficult
- “Gutenberg” ranking on Alice’s Adventure in Wonderland could be an error from the text including the website’s add ins - what other words could have their rank effected?

Results


Alice's Adventures in Wonderland
said
alice
little
one
project
went
like
would
could
thought
gutenberg
see
queen
know
king
0
100
200
300
400
Through the Looking-Glass
said
alice
queen
one
like
little
it's
project
went
thought
know
would
see
don't
red
0
100
200
300
400

Results


Alice's Adventures in Wonderland
said
alice
little
one
project
went
like
would
could
thought
gutenberg
see
queen
know
king
0
100
200
300
400
Dracula
could
one
said
must
us
would
shall
may
know
see
time
van
come
came
went
0
100
200
300
400
500

Results

- Lewis Carroll uses more names, similar words throughout both books, and clear frequency drop off
- Bram Stoker uses a more even distribution, and frequent words are more “basic” English supporting vocabulary

New Ideas

- Could word frequency be used to determine how descriptive or “basic” a text is?
- Further, could this be used to classify the difficulty of reading for non native speakers, such as assigning a CEFR level?